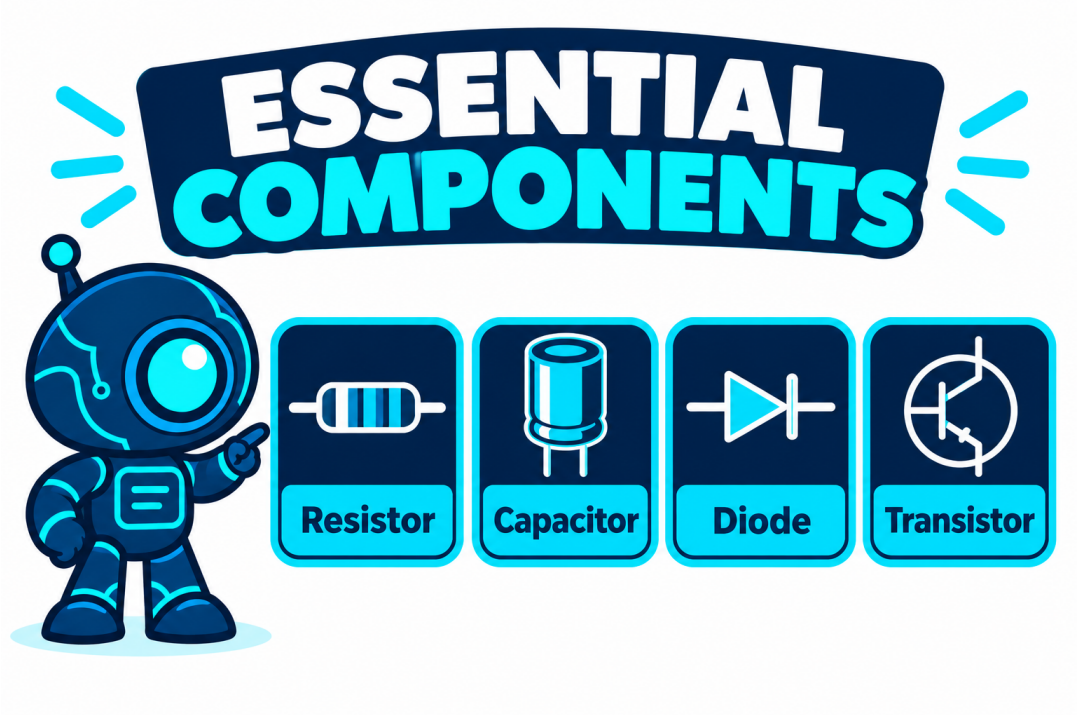
AI MECH LAB

Mechatronics Course — Beginner Track



Module 2, Topic 2.3

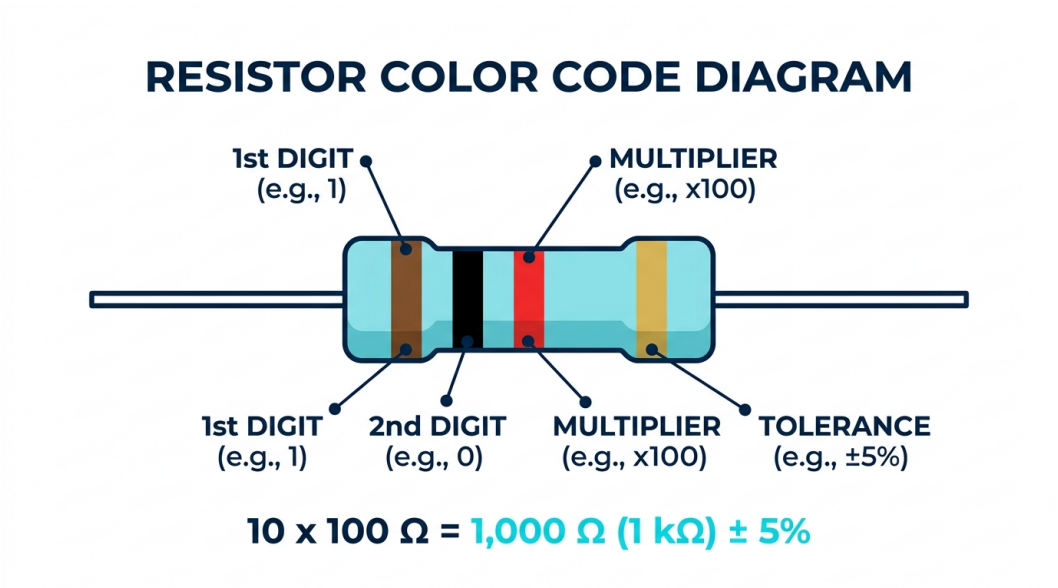
Essential Components

Lesson

You now understand how components connect — series and parallel — and the three quantities that describe what's happening in a circuit. This lesson introduces the components themselves: four parts that show up in almost every circuit you'll ever build. You've already met one of them. The other three are new, and each does something the other three can't.

Resistors — controlling current

You already know what a resistor does from Topic 2.1: it adds a known, controlled amount of resistance to a circuit, limiting current to a safe level. Resistors are color-coded with bands that tell you their resistance value — a skill worth building early, since you won't always have a multimeter handy to check.

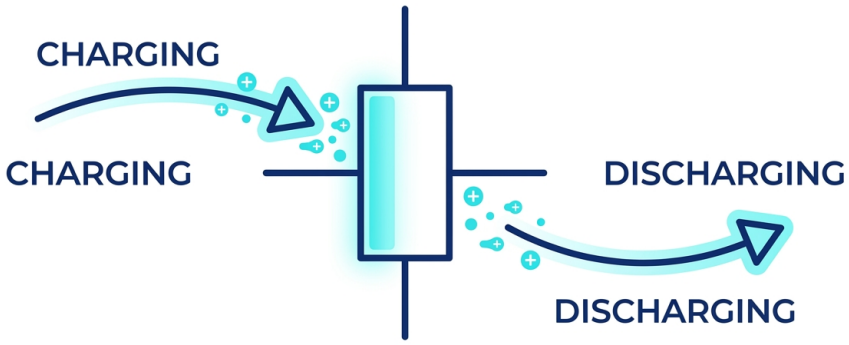


Color bands on a resistor encode its resistance value and tolerance.

Resistors are non-polarized, meaning they work the same way no matter which direction current flows through them. You can install one backwards and it behaves identically. That's not true for every component on this list.

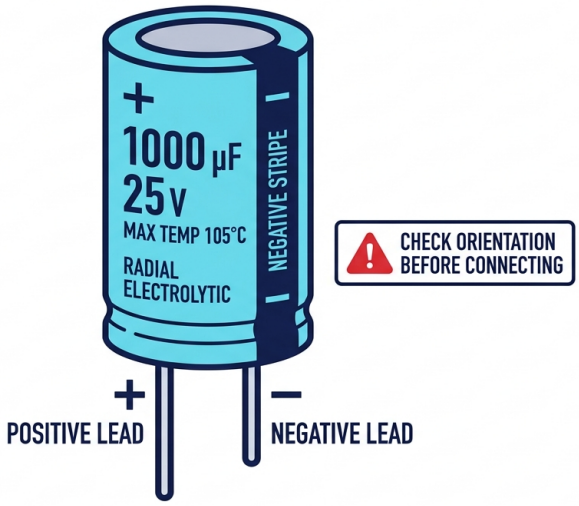
Capacitors — storing and releasing charge

A capacitor stores electrical charge temporarily, then releases it when needed. Picture it like a small, fast-charging reservoir: it fills up with charge while voltage is applied, and discharges that stored charge back into the circuit when voltage drops or the capacitor is connected across a path.



A capacitor filling with charge, then releasing it back into the circuit.

This storage behavior makes capacitors useful in a few recurring roles. They smooth out power supply fluctuations by filling in small dips in voltage. They filter unwanted electrical noise out of a signal. And they can introduce a brief time delay in a circuit, since charging or discharging isn't instant — it takes a small, predictable amount of time.

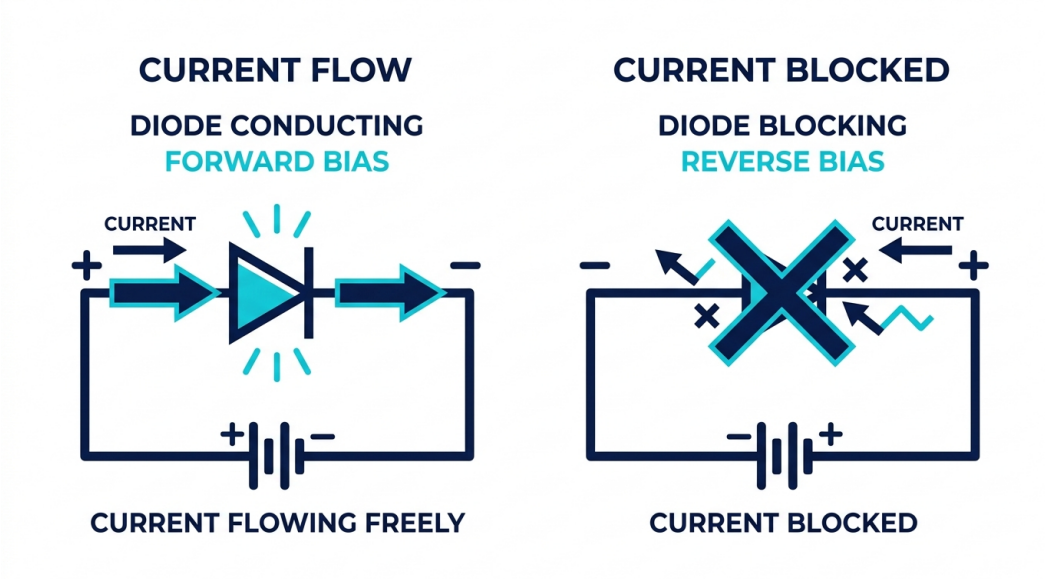


A polarized capacitor's positive and negative leads matter.

Some capacitors are polarized, meaning they have a distinct positive and negative terminal, and connecting one backwards can damage the component or the circuit it's in. Always check before wiring one in — polarized capacitors are usually clearly marked with a stripe or a longer lead on the positive side.

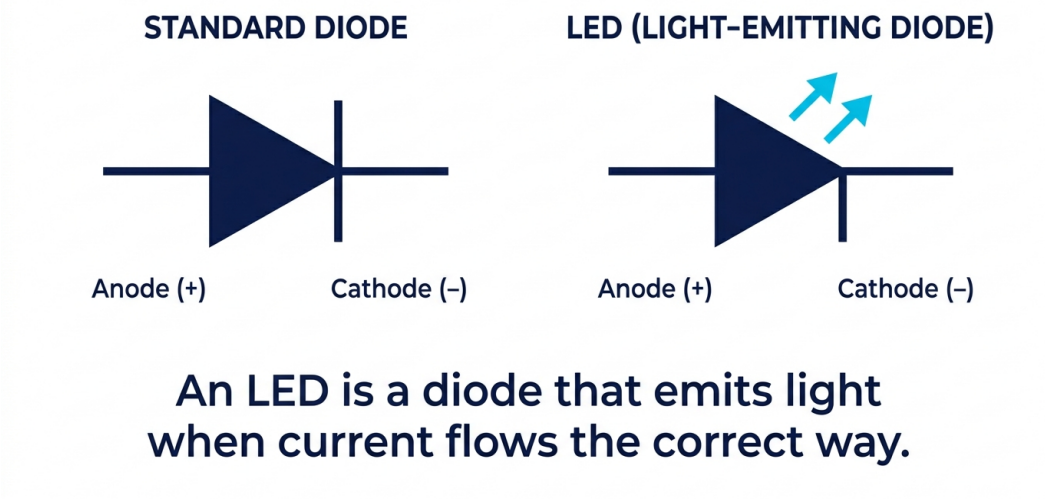
Diodes — one-way valves for current

A diode allows current to flow in only one direction and blocks it in the other. Think of it as a one-way valve for electricity: current flows freely one way, and hits a wall going the other way.



Current flows freely one direction through a diode, and is blocked the other.

This makes diodes essential for protecting circuits from reversed connections — if someone (including you, on a bad day) connects a battery backwards, a correctly placed diode can prevent current from flowing the wrong way through sensitive components, protecting them from damage.



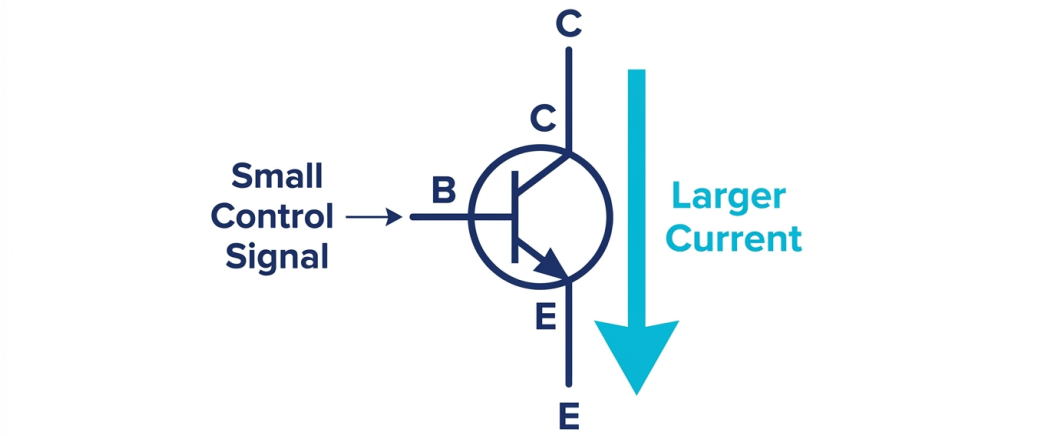
An LED is a diode that happens to emit light when current flows the correct way.

Diodes are also the basis of how an LED works: an LED is simply a diode that happens to emit light when current flows through it in the correct direction, which is exactly why LEDs only light up one way around.

Diodes are always polarized — direction matters completely. Install one backwards, and at best it simply blocks current entirely; at worst, depending on the circuit, it can fail or damage other components.

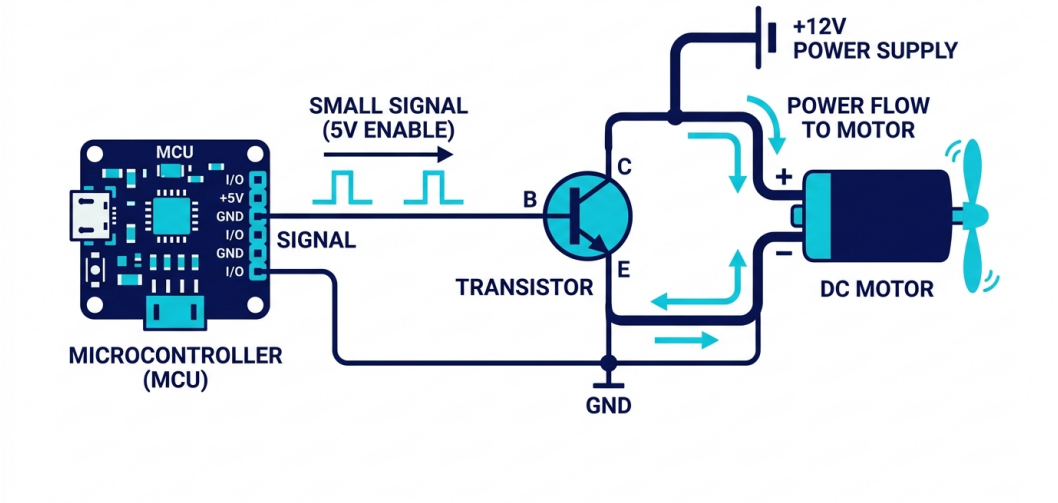
Transistors — the switch that controls itself

A transistor acts as an electronically controlled switch, but with an important difference from a simple mechanical switch: instead of a person flipping it, a small signal controls whether a much larger current is allowed to flow. A tiny current or voltage on one terminal determines whether the transistor allows current to pass between its other two terminals.



A small control signal determines whether a much larger current flows.

This is the property that makes transistors so powerful: a small signal — like the output from a microcontroller — can control something requiring far more current than the microcontroller could ever supply directly, such as a motor or a bright light. The microcontroller doesn't power the motor itself; it tells the transistor to allow power to the motor.



A microcontroller's small signal telling a transistor to power a motor.

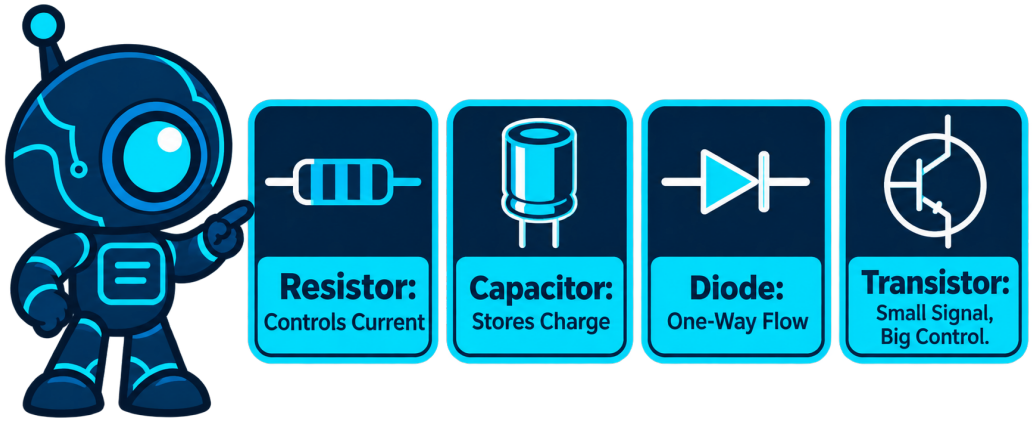
Transistors have three terminals, and getting the right terminal connected to the right part of the circuit matters — unlike a resistor, you can't just connect it any way and expect it to work.

Why these four, specifically

These four components recur constantly because each one does something fundamentally different, and most real circuits need some combination of all four: resistors to control current levels, capacitors to store charge and smooth behavior over time, diodes to enforce one-way current flow, and transistors to let a small signal control something larger. Once you start reading circuit diagrams for sensors and motors later in this course, you'll recognize these four everywhere — often working together in the same small section of a circuit.

Where this leaves you

By the end of this lesson, you should be able to name what each of these four components does, recognize which ones are polarized and need correct orientation, and explain in plain terms why a transistor is fundamentally different from the other three — it's the one component here that lets a small signal control something much larger. Next, you'll put these ideas to direct practical use: reading a component's datasheet and using a multimeter to verify what's actually happening in a real circuit.



Four components, four distinct jobs in every circuit.